Code No: 113AU **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, May/June - 2019
ELECTRONIC DEVICES AND CIRCUITS
(Common to EEE, ECE, CSE, EIE, IT, MCT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

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PART- A

(25 Marks)

1.a) Find the value of dc resistance of a Ge junction diode, if the temperature is 25^0 C and I_0 = 20μA with an applied voltage of 0.1V. [2]
b) Briefly discuss the semiconductor photo diode. [3]
c) What are the limitations of zener voltage regulator circuit? [2]
d) Compare half wave rectifier and full wave rectifier. [3]
e) What is Early Effect? [2]
f) Given an NPN transistor for which α_N = 0.98, I_{CO} = 2mA. A common emitter connection is used and V_{CC} = 12V, R_L = 4.0 KΩ. What is the minimum base current required in order that the transistor enter its saturation region. [3]
g) Define thermal stability. [2]
h) Define Thermal Runaway and Thermal Resistance. [3]
i) What are applications of JFET and MOSFET devices? [2]
j) Explain how FET acts as a voltage regulator. [3]

PART-B

(50 Marks)

2.a) Derive the expression for contact difference of potential V_0 in an open circuited p-n junction.
b) Explain Transition capacitance and Diffusion capacitances. [5+5]

OR

3.a) Explain the principle operation of varactor diode.
b) Write the differences between Tunnel Diode and PN Diode. [5+5]

4.a) Obtain the expression for ripple factor in the case of Half Wave Rectifier with π filter.
b) A diode whose internal resistance is 20 Ω is to supply power to a 1000 Ω load from a 110V (rms) source of supply. Calculate:
i) The peak load current
ii) The DC load current
iii) AC load current
iv) The DC diode voltage
v) The total input power to the circuit
vi) % regulation from no load to the given load. Assume Half wave rectifier. [5+5]

OR

5.a) Explain the working of a bridge rectifier with a neat circuit diagram and with relevant wave forms.
b) Compare C, L, L-section and π-section Filters in all respects. [5+5]

6.a) Explain the construction and working of BJT.

b) In a CB configuration current amplification factor is 0.90 and emitter current is 1mA. Determine base current. [8+2]

OR

7.a) Describe the V-I characteristics of a transistor in common collector configuration and explain.

b) What is the significance of negative resistance region? Explain how UJT exhibits this characteristics? [6+4]

8.a) Draw the fixed bias circuit and derive the expression for the stability factor S. What are the limitations of this circuit?

b) In a voltage divider bias circuit, V_{CC} = 20 V and R_C = 1.5 kΩ, the Q point is V_{CE} = 8V and I_C = 4mA. Stability factor S = 12 and β = 50. Find R_1, R_2 and R_E. [5+5]

OR

9. Derive the expressions for all gains and Impedances of a CE Amplifier using h-parameters. [10]

10.a) Obtain the expression for the pinch off voltage V_P in the case of n-channel JFET.

b) Draw the structure of p-channel MOSFET and qualitatively explain the static drain and gate characteristics of the device. [5+5]

OR

11.a) Derive an expression for voltage gain, Input Impedance and output impedance of CG amplifier at low frequencies.

b) For the emitter follower with R_S = 0.5KΩ and R_L = 5KΩ, calculate A_I, R_i, A_{VS} and R_0. Assume h_{fe}=50, h_{ie}=1KΩ, h_{oe}=25μA/V. [6+4]

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